Alex Nham

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Education

The University Of Western Ontario — London, ON
B.Sc. in Computer Science, Minor in Software Engineering (GPA 3.85) — *Expected Graduation May 2027*

Experience

Software Engineer Intern — Sept 2026 – Dec 2026
PointClickCare — *Toronto, ON*
- Developing Soundcheck, a post-deployment application maturity platform using React, TypeScript, Terraform, Kubernetes and REST APIs to assess production health and operational readiness
- Building automated health checks and monitoring dashboards to identify deployment issues and improve application reliability

Software Engineer Intern — May 2025 – Sept 2026
Celestica — *Toronto, ON*
- Accelerated internal issue resolution by building an **AI support tool** with TypeScript, Gemini API, LangChain, ChromaDB, and Docker, implementing a **RAG pipeline** over historical support data to surface relevant solutions instantly
- Built **real-time SPC** systems in C# and .NET with ingestion pipelines, control chart calculations, alerting, and reporting, giving **150+ engineers** across global production lines visibility into process deviations
- Built and maintained full-stack .NET internal engineering tools and dashboards across a **hybrid on-prem, GCP, and Azure** architecture, keeping sensitive production data on-prem while leveraging cloud services for compute and analytics
- Reduced workflow execution time by **50%** by automating critical operational processes with Python, Bash, and PowerShell scripts, eliminating repetitive manual tasks and improving engineering team productivity

Software Engineer Intern — April 2024 – Sept 2024
AQX AI — *New York City, NY*
- Engineered a high-performance Node.js backend for **real-time AI agent communication**, reducing end-to-end latency by **90%** through optimized WebSocket handling, asynchronous message queuing, and connection pooling
- Built the primary React.js operator interface for managing and initiating outbound AI agent calls, serving as the sole client for agent interaction and enabling non-technical operators to deploy and monitor calls in real time
- Containerized the full-stack application and databases using Docker and Kubernetes, automating CI/CD deployment pipelines on Azure cloud infrastructure to ensure consistent cross-environment delivery and streamlined production releases

Website Coordinator — August 2022 – April 2024
Talbot Promotions — *London, ON*
- Designed and maintained e-commerce storefronts for **30+ companies**, delivering polished UX for corporate merchandise sales
- Built Python scripts integrating external APIs, streamlining workflows and reducing manual processing time by **70%**

Projects

Coin Operated Module | *C, TypeScript, Next.js, PostgreSQL, Stripe* — 2026
- Engineered a full-stack payment platform and custom hardware module that modernizes coin-operated laundry machines, enabling residents to pay and remotely start machines, processing **$600/month** across **20+ users**
- Integrated Stripe with webhooks and transaction workflows to ensure reliable payment processing and credit allocation
- Interfaced with an **ESP32 microcontroller** and relay module via database polling, triggering real-time hardware actions and seamlessly synchronizing physical machine state with user credit transactions

AI Audio Summarizer | *Angular, Node.js, Express, Deepgram, OpenAI, Supabase* — 2025
- Eliminated hours of manual audio review by building a full-stack AI pipeline using Deepgram and OpenAI to automatically transcribe and summarize long-form audio, reducing processing time from hours to seconds
- Architected chunked audio handling with Vercel Blob Storage to process long-form files that exceeded single-request limits, with Supabase for persistent storage and OAuth 2.0 authentication

Adrenal Nhamogram | *Flutter, React.js, Python* — 2024
- Co-created a **validated clinical scoring system** for adrenal incidentalomas, presented at the **AUA Annual Meeting 2024**, enhancing diagnostic accuracy and supporting data-driven clinical decision-making for urologists
- Delivered the scoring tool across iOS, Android, and web by building a Flutter mobile app and React.js web app, maximizing accessibility for healthcare professionals across all devices

Open Source Contribution

Sprocket (St. Jude Children's Research Hospital) | *Rust* — 2026
- Contributed to St. Jude's open-source Rust WDL workflow engine (Sprocket), implementing lint rules and fixing formatter and parsing issues to improve reliability and enforce workflow best practices for clinical bioinformatics pipelines

Technical Skills

Languages: Python, TypeScript, Java, C++, C#, JavaScript, SQL, Rust, C, Bash
Frameworks & Libraries: React.js, Node.js, Express.js, Flask, Django, .NET, React Native, Flutter, Tailwind CSS
Databases: MySQL, MongoDB, PostgreSQL, Microsoft SQL Server
Cloud & DevOps: Azure, GCP, AWS, Docker, Kubernetes, GitHub Actions, CI/CD Pipelines, RESTful APIs, Postman, Nginx
Tools & Platforms: Git, Visual Studio Code, Figma, Jira, Unix/Linux
Additional Skills: System Administration, Automation Scripting, Code Review, Performance Optimization, Technical Documentation